

But, what if we could extend this automation to the individual level to archive medical histories and track recoveries? What if disease tracking went beyond the walls of the physical hospitals?

Today we have tons of fitness applications that track a patient's pulse, BMR, diet, exercise, sleep, water intake, and other vital biological statistics. These applications integrate smart watch sensor readings with mobile widgets to populate user details. Automation can work along similar lines, integrating patient statistics with pre-populated medical data repositories. It can facilitate proactive patient follow-ups, monitor disease progression and holistically enhance the quality of medical services. When integrated with bio-medical sensors of smart devices, it can provide useful physiological insights to medical practitioners.

Centralised Web based disease surveillance is the need of the hour for active disease tracking of individual patients. Imagine if all doctors had access to the medical history of their patients at a click of a button, without having to manually review each of the paper files or verbal declarations. Would it not accelerate the time taken for treatment and the overall quality of medical services? Centralised disease surveillance systems are also some of the fundamental precursors to implement successful telemedicine services.

The technology that we need in order to create advanced and holistic medical automation systems already exists today. We just need someone to put together all the different aspects to create the right product.

Machine learning for disease surveillance

The success of the PMO's Aarogya Sethu mobile application can be primarily attributed to the

government's interest in solving human problems using technology. During this COVID-19 pandemic, contact tracing and disease tracking was done using the Blue Light Energy (BLE) functionality of mobile devices. Comprehensive information on infected citizens, and all those who came in proximity with them, was sent to the central server. Citizen alerts with options for preventive actions were sent to the users.

Pandemic or disease spread has a spatial and socio-demographical context. COVID-19 fatalities, for instance, are more prominent in the elderly population than other age groups. Now, what if we had software that is able to infer such trends and also pre-populate a cadastral hotspot map, with all the contact and address details of the identified elders in a specified age group? Gaining insights into such trends can help the government take appropriate disaster mitigation steps in real-time. Machine learning has the power to provide such robust insights. Not only can the deep learning algorithms predict trends, but they can also generate spatial maps and tabular reports of all vulnerable populations. These outputs can significantly help the government, epidemiologists, health care workers and other stakeholders make informed decisions that can efficiently control the disease's spread.

Artificial intelligence and machine learning can also help enforcement officials monitor citizen compliance with public policies. Deep learning algorithms can compute the dynamic relative coordinates of moving objects or pedestrians, and even detect anomalies between them. All that is needed is a bird's eye view input data

set of the target objects. The street surveillance cameras, which serve as a bird's eye view data set, can be integrated with Python libraries to detect pedestrian violations.

Many countries have used AI, machine learning, and the Internet of Things to monitor social distancing among citizens. The systems are able to measure the actual distances between individual pedestrians, in units of metres or kilometres, by computing the sum of relative pixel distances between them. Such AI based applications are necessary for public enforcement monitoring, and controlling disease spread.

Artificial intelligence (AI), machine learning, the Internet of Things (IoT) and Web applications have enhanced the quality of our lives. Until now, a large section of society resisted automation in the fear that AI was taking over humans. But if these advancements had already been in place, AI could have saved many human lives and better controlled the COVID-19 pandemic. We would not have to wait long in queues just to get tested, nor would the expert healthcare practitioners have had to work overtime. The AI-driven systems approach and applications could have provided the necessary technology and insights to better manage the pandemic.

AI will not take over humans. Instead, it will support them. Humans will be able to perform even better at their jobs by working alongside AI-driven systems. AI has the power to control any epidemic, not just COVID-19. Hence, in the better interest of society, let us embrace the power of AI. 

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